

Problem 1 (AIME II 2019/6)

We let $u = \log(x)$.

$$3 \log(\sqrt{x} \log(x)) = 56 \Leftrightarrow 3 \left(\frac{1}{2} u + \log(u) \right) = 56$$

$$\log_{\log(x)}(x) = \frac{\log(x)}{\log(\log(x))} = 54 \Leftrightarrow \frac{u}{\log(u)} = 54 \Leftrightarrow u = 54 \log(u) \Leftrightarrow b^u = u^{54}$$

Plugging this back into,

$$3(27 \log(u) + \log(u)) = 56 \Leftrightarrow \log(u) = \frac{2}{3} \Leftrightarrow u = b^{\frac{2}{3}} \Leftrightarrow u^{54} = b^{36}$$

Therefore, $u = 36$, so $b = 36^{\frac{2}{3}} = 216$.

Problem 3 (CEMC Euclid 2008/9C)

We plug in $x = 10^a$; $y = 10^b + 1$; $z = 10^c + 2$ to get,

$$10^a + a = 10^b;$$

$$10^b + b = 10^c;$$

$$10^c + c = 10^a$$

If $a \neq 0$, we can get that either $a < b < c < a$ or $a > b > c > a$ to get a contradiction. Therefore, $a = b = c = 0$ and the only solution is $(x, y, z) = (1, 2, 3)$.

Problem 4 (CMIMC 2018 A5)

Notice that,

$$bc + \frac{1}{a} = \frac{1}{a+b+c} \Leftrightarrow abc = \frac{a}{a+b+c} - 1$$

Similarly,

$$abc = \frac{b}{a+b+c} - 2 = \frac{c}{a+b+c} - 7$$

Adding these up, we get,

$$3abc = -10 + \frac{a+b+c}{a+b+c} \Leftrightarrow abc = -3$$

Now, we multiply the equations together to get,

$$\left(bc + \frac{1}{a}\right)\left(ca + \frac{2}{b}\right)\left(ab + \frac{7}{c}\right) = (abc)^2 + 10abc + 23 + \frac{14}{abc} = 9 - 30 + 23 + \frac{14}{-3} = \frac{1}{(a+b+c)^3}$$

Therefore, $a + b + c = -\frac{\sqrt[3]{3}}{2}$.

6 clubs; 5 hours (including examples)

Problem 6 (EGMO 2019/1)

We claim the only solutions are the permutations of $(a, b, c) = (0, 1, 1)$ and $(a, b, c) = (0, -1, -1)$, and also $(a, b, c) = \left(\frac{\sqrt{3}}{3}, \frac{\sqrt{3}}{3}, \frac{\sqrt{3}}{3}\right)$ or $(a, b, c) = \left(-\frac{\sqrt{3}}{3}, -\frac{\sqrt{3}}{3}, -\frac{\sqrt{3}}{3}\right)$. Each of these obviously works.

Using $ab + bc + ca = 1$,

$$a^2b + c = b^2c + a = c^2a + b$$

$$\Leftrightarrow a^2b + abc + bc^2 + ac^2 = b^2c + a^2b + abc + a^2c = ac^2 + ab^2 + b^2c + abc$$

Taking the first equality, we cancel to get,

$$bc^2 + ac^2 = b^2c + a^2c$$

For $c = 0$, we get $ab = 1$; $a^2b = a = b$ so $a = b = \pm 1$.

This gives the permutations of $(a, b, c) = (0, 1, 1)$ and $(a, b, c) = (0, -1, -1)$. If not,

$$\Rightarrow bc + ac = b^2 + a^2$$

Similarly, we get,

$$ab + ac = b^2 + c^2;$$

$$ab + bc = a^2 + c^2$$

Adding these up, we get,

$$\Leftrightarrow 0 = (a - b)^2 + (b - c)^2 + (c - a)^2$$

Therefore, $a = b = c$. From the given, we get the last solution, where $(a, b, c) = \left(\frac{\sqrt{3}}{3}, \frac{\sqrt{3}}{3}, \frac{\sqrt{3}}{3}\right)$ and $(a, b, c) = \left(-\frac{\sqrt{3}}{3}, -\frac{\sqrt{3}}{3}, -\frac{\sqrt{3}}{3}\right)$.

Problem 7 (OMC 2013 F24)

Note that,

$$(a_n - a_{n-1})(b_n + b_{n-1}) = \frac{2n+2}{6048} = \frac{n+1}{3024}$$

$$\Leftrightarrow a_nb_{n-1} - a_{n-1}b_n = \frac{n+1}{3024} - a_nb_n + a_{n-1}b_{n-1}$$

Therefore,

$$\begin{aligned}\sum_{k=1}^{2013} (a_k a_{k-1} - a_{k-1} b_k) &= \frac{1}{3024} \cdot \left(\sum_{k=1}^{2013} (k+1) + \sum_{k=0}^{2012} a_n b_n - \sum_{k=1}^{2013} a_n b_n \right) \\ &= \frac{1}{3024} \cdot \left(\frac{2013 \cdot 2016}{2} + a_0 b_0 - a_{2013} b_{2013} \right) \\ &= 671\end{aligned}$$

12 clubs; 6.5 hours (including examples)

Problem 8 (Vietnam 2014/1)

We claim that $x_k = \sin\left(\frac{60}{2^k}\right)$; $y_k = \cos\left(\frac{60}{2^k}\right)$. We prove this by induction,

Base case: $k = 1$, already given.

Inductive step: $k = n$ to $k = n + 1$,

$$\begin{aligned}x_{n+1}^2 + y_n &= 2 \\ \Leftrightarrow x_{n+1}^2 + 2 \cos\left(\frac{60}{2^n}\right) &= 2 \\ \Leftrightarrow x_{n+1}^2 &= 2 - \left(2 - 4 \sin^2 \frac{60}{2^{n+1}}\right) = 4 \sin^2 \frac{60}{2^{n+1}}\end{aligned}$$

Since x_{n+1} is positive,

$$\Leftrightarrow x_{n+1} = 2 \sin\left(\frac{60}{2^{n+1}}\right)$$

Now,

$$\begin{aligned}x_{n+1} y_{n+1} - x_n &= 0 \\ \Leftrightarrow 2 \sin\left(\frac{60}{2^{n+1}}\right) y_{n+1} &= 2 \sin\left(\frac{60}{2^n}\right) \\ \Leftrightarrow 2 \sin\left(\frac{60}{2^{n+1}}\right) y_{n+1} &= 4 \sin\left(\frac{60}{2^{n+1}}\right) \cos\left(\frac{60}{2^{n+1}}\right)\end{aligned}$$

Therefore, $y_{n+1} = 2 \cos\left(\frac{60}{2^{n+1}}\right)$ as well.

Conclusion: Induction complete for all $k \geq 1$.

Therefore,

$$\lim_{n \rightarrow \infty} x_n = 0; \lim_{n \rightarrow \infty} y_n = 2$$

Problem 9 (IMO 2014/1)

For the sake of contradiction, let none of the expressions be true. We multiply n to each equation to get,

$$a_1 < a_0 + a_1 \leq a_2;$$

$$2a_2 < a_0 + a_1 + a_2 \leq 2a_3;$$

$$3a_3 < a_0 + a_1 + a_2 + a_3 \leq 3a_4;$$

...

We prove by induction that for all $n \geq 1$, $na_{n+1} < a_n + \dots + a_0$.

Base case: $n = 1$,

The equation $a_1 < a_0 + a_1$ is true since $a_0 > 0$, so, in the first expression, the second inequality must be false, so $a_0 + a_1 > a_2$.

Inductive step: $n = k$ to $n = k + 1$,

We know that at least one of the inequalities,

$$(k + 1)a_{k+1} < a_0 + \dots + a_{k+1} \leq (k + 1)a_{k+2}$$

We also know,

$$a_0 + a_1 + \dots + a_k > ka_{k+1} \Leftrightarrow (k + 1)a_{k+1} < a_0 + \dots + a_{k+1}$$

Therefore, the other inequality must be false as desired.

Conclusion: By induction the statement is true.

We therefore get,

$$a_2 < a_0 + a_1;$$

$$2a_3 < a_0 + a_1 + a_2;$$

...

Adding up the first k equations,

$$a_2 + 2a_3 + 3a_4 + \dots + ka_{k+1} < ka_0 + ka_1 + (k - 1)a_2 + (k - 2)a_3 + \dots + a_k$$

As k approaches infinity, the LHS will become greater than the RHS as a is increasing over integers. We're hence done since there's a contradiction. (Can you check if this statement is ok w/ graders?)

17 clubs; 8 hours (including examples)

Problem 10 (2014 AIME/14)

Sketch (wrote it all out on OneNote and deleted it ;-).

Move -4 over and add 1 to each fraction, then divide by x .

Substitute $y = x - 11$, $z = y^2 - 50$ to get apart from $x = 0, y = 0$, we also have two more solutions for z .

Maximal solution ends up being $x = 11 + \sqrt{50 + 10\sqrt{2}}$.

Problem 11 (Z46CD2B6)

Lemma: $f(x) = x + \sqrt{1 + x^2}$ is positive and injective.

Note that, $\sqrt{1 + x^2} > |x| \geq -x$. We immediately see that the function is positive. We also have,

$$\begin{aligned} \Rightarrow f'(x) &= 1 + \frac{x}{\sqrt{1 + x^2}} \\ \sqrt{1 + x^2} > -x &\Rightarrow \frac{x}{\sqrt{1 + x^2}} > -1 \Rightarrow 1 + \frac{x}{\sqrt{1 + x^2}} > 0 \end{aligned}$$

Since f is strictly increasing, f is injective.

We claim that $x + y = 0$.

For the lemma, $y + \sqrt{1 + y^2} \neq 0$, so we have,

$$\begin{aligned} (x + \sqrt{1 + x^2})(y + \sqrt{1 + y^2}) &= 1 \\ \Leftrightarrow x + \sqrt{1 + x^2} &= \sqrt{1 + y^2} - y \end{aligned}$$

We plug in $Y = -y$,

$$\begin{aligned} \Leftrightarrow x + \sqrt{1 + x^2} &= \sqrt{1 + Y^2} + Y \\ \Leftrightarrow x &= Y \\ \Leftrightarrow x + y &= 0 \end{aligned}$$

Problem 12 (EGMO 2015/4)

We claim the answer is $n = 5$, which can be constructed with $(a_1, a_2, a_3, a_4, a_5) = (477, 7, 29, 35, 43)$.

We let $x_n = a_{n+1} - a_n$. We know x_n is always positive for $n \geq 2$ since square roots are positive. Since a is increasing, x_n is also increasing for $n \geq 3$.

Notice that,

$$x_{n+1}^2 - x_n^2 = a_{n+1} - a_{n-1} = x_n + x_{n-1}$$

$$\Leftrightarrow x_{n+1} - x_n = \frac{x_n + x_{n-1}}{x_{n+1} + x_n}$$

For $n \geq 4$, the RHS is less than 1. However, $x_{n+1} - x_n \geq 1$, so there's a contradiction. Hence, the maximum length of the sequence is 5.

Problem 13 (AIME II 2006/15)

Note that $x, y, z > 0$. In addition, $x = \sqrt{y^2 - \frac{1}{16}} + \sqrt{z^2 - \frac{1}{16}} < y + z$ and similar results means that x, y, z can be the side lengths of a triangle.

By the Pythagorean Theorem, we can have x, y, z be the side lengths of a triangle with altitudes of length $\frac{1}{4}$ to x , $\frac{1}{5}$ to y , and $\frac{1}{6}$ to z .

Using $A = bh$, we can write $x = 4k, y = 5k, z = 6k$. We use Heron's to get,

$$A = \frac{15k^2\sqrt{7}}{4} = \frac{k}{2} \Leftrightarrow k = \frac{2\sqrt{7}}{105}$$

Therefore, $x = \frac{8\sqrt{7}}{105}; y = \frac{2\sqrt{7}}{21}; z = \frac{4\sqrt{7}}{35}$.

30 clubs; 11 hours (including examples)

Problem 14 (Baltic Way 2020)

We claim that the only solution is $x = y = z = 0$, which obviously works.

We can write the first equation using the quadratic formula,

$$x = \frac{\pm\sqrt{-4y^3z - 4yz^2}}{2y};$$

$$y = \frac{-x^2 \pm \sqrt{x^4 - 4z^3}}{2z};$$

$$z = \frac{-y^2 \pm \sqrt{y^4 - 4x^2y}}{2}$$

For the solution to be real, all the discriminants must be non-negative,

$$-4y^3z - 4yz^2 \geq 0;$$

$$x^4 - 4z^3 \geq 0;$$

$$y^4 - 4x^2y \geq 0$$

Adding the equations up and dividing by 4, we get the second equation. Therefore, all the inequalities must be equalities. It is impossible for x, y, z to have any other solutions. (abbreviated)

Problem 16 (Mathematic Reflections J479)

We let,

$$x = \frac{a^2}{bc} - 1; y = \frac{b^2}{ac} - 1; z = \frac{c^2}{ab} - 1$$

Therefore,

$$\begin{aligned} &\Leftrightarrow x^3 + y^3 + z^3 = 3xyz \\ &\Leftrightarrow (x + y + z)(x^2 + y^2 + z^2 - xy - yz - zx) = 0 \end{aligned}$$

Case one: $x + y + z = 0$, meaning,

$$\begin{aligned} &\frac{a^2}{bc} + \frac{b^2}{ac} + \frac{c^2}{ab} = 3 \\ &\Leftrightarrow (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca) = 0 \end{aligned}$$

Note that,

$$\begin{aligned} &a^2 + b^2 + c^2 - ab - bc - ca = 0 \\ &\Leftrightarrow \frac{(a-b)^2}{2} + \frac{(b-c)^2}{2} + \frac{(c-a)^2}{2} = 0 \end{aligned}$$

We get, $a = b = c$ which is not allowed.

Therefore, we must have $a + b + c = 0$ in this case.

Case two: $x^2 + y^2 + z^2 - xy - yz - xz = 0$

In this case, we similarly end up with $x = y = z$, so,

$$\begin{aligned} &\frac{a^2}{bc} - 1 = \frac{b^2}{ac} - 1 = \frac{c^2}{ab} - 1 \\ &\Leftrightarrow a^3 = b^3 = c^3 \\ &\Leftrightarrow a = b = c \end{aligned}$$

Once again, this is not allowed, so this case is impossible.

The only possibility results in $a + b + c = 0$.

36 clubs; 13 hours

Problem 25 (IMO 2018/2)

We claim the answer is all the multiples of 3, which can be achieved with the sequence -1, -1, 2, -1, -1, 2, ...

To prove that all other n fail, note that,

$$\begin{aligned} a_i a_{i+1} + 1 = a_{i+2} &\Leftrightarrow a_i a_{i+1} a_{i+2} = a_{i+2}^2 - a_{i+2}; \\ a_{i+1} a_{i+2} + 1 = a_{i+3} &\Leftrightarrow a_i a_{i+1} a_{i+2} = a_i a_{i+3} - a_i \end{aligned}$$

Summing $a_{i+2}^2 - a_{i+2} = a_i a_{i+3} - a_i$ cyclically, we get,

$$\sum_{cyc} a_{i+2}^2 - a_{i+2} = \sum_{cyc} a_i a_{i+3} - a_i$$

$$\Leftrightarrow a_1^2 + a_2^2 + \dots + a_n^2 = a_1 a_4 + a_2 a_5 + \dots + a_{n-2} a_1 + a_{n-1} a_2 + a_n a_3$$

If n is not a multiple of 3, we take indices mod 3 to get,

$$\Leftrightarrow a_1^2 - a_1 a_4 + a_4^2 - a_4 a_7 + a_7^2 + \dots = (a_1 - a_4)^2 + (a_4 - a_7)^2 + \dots = 0$$

We end up getting $a_1 = a_2 = \dots = a_n$. We solve $x^2 + 1 = x$ to get there's no solution.

Problem 27 (Iberoamerican 2021/4)

WLOG, let $1 = a^2 + x^2 = \dots$. We let, complex numbers,

$$z_1 = a + xi; z_2 = b + yi; z_3 = c + zi$$

We have,

$$1 = |z_1| = |z_2| = |z_3| = |z_1 + z_2| = |z_2 + z_3| = |z_3 + z_1|$$

If we draw the unit circle ω_1 on the complex plane and another unit circle ω_2 centered at z_1 . We let z_3, z_4 be the intersections of ω_1, ω_2 , we see that $z_2 = z_3 - z_1$ or $z_2 = z_4 - z_1$, so z_2 is 120 degrees clockwise or counter-clockwise from z_1 .

The same applies to all the others, so z_1, z_2, z_3 are all in 120 degree apart. Therefore, so are z_1^2, z_2^2, z_3^2 .

$$z_1^2 + z_2^2 + z_3^2 = 0$$

$$\Rightarrow \text{Re}(z_1^2 + z_2^2 + z_3^2) = 0$$

$$\Leftrightarrow a^2 + b^2 + c^2 = x^2 + y^2 + z^2$$